Akbar Aman

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Summary

Computer Engineer experienced in designing enterprise AI systems, digital infrastructure, production software, AI evaluation & governance, ML research, & hardware/software co-design.

Education

University of Illinois Chicago, College of Engineering Expected: May 2027

B.S. in Computer Engineering (Computer & Networked Systems Track), Minor in Mathematics *GPA: 3.54*

- **Relevant Coursework:** Comp Arch, Networking, Data Structures, Systems Programming, DSP, HDL, VLSI Design
- **Leadership:** Computer Science Teaching Assistant (MATLAB, 2 terms) and Engineering Success Mentor (4 terms).

Experience

Enterprise AI Intern May 2026 – Present

AHEAD *Hybrid — Chicago, IL*

- Promoted from a summer internship into a year-round role on AHEAD's Enterprise GPT team, supporting internal R&D, engineering, & deployment while taking technical ownership of an AI skills & agent governance initiative for Glean & Claude. Developed a validated self-service model for creation, discovery, review, & reuse across licensed users.
- Authored the foundational narrative for a proprietary agentic platform, translating its architecture, capabilities, and operating model into a CTO-cleared source of truth for stakeholder approval, internal enablement, & GTM strategy.

AI Trainer, Model Evaluation & Safety Apr 2026 – Present

LinkedIn *Multiple Engagements*

- Professional Services (Sep 2026 – Present; via AKA Services): Invited to a specialized domain-expert engagement developing expert-level software development & information-systems tasks for frontier-model training & evaluation.
- AI Safety / Red Teaming (Apr 2026 – Aug 2026): Red-teamed frontier LLMs through adversarial prompting & structured evaluations, uncovering and characterizing failure modes across reasoning, coding, & instruction-following.

AI-EDGE REU Fellow Jun 2026 – Jul 2026

NSF AI-EDGE Institute, The Ohio State University *Remote*

- Selected for the NSF AI-EDGE Summer program at OSU, completing advanced study across LLMs, generative models, AI security, and distributed AI, with applied work in Vision Transformers, agent security, and multi-agent systems.

AI Systems Engineer Oct 2025 – Mar 2026

Talent Strategy Experts *Remote*

- Architected & implemented the minimum viable product of a multi-tenant SaaS platform for AI inference, owning end-to-end fullstack, deployment, system documentation, & preparing the system for team-based development & scale. Designed isolation boundaries, role-based access control, and reliability safeguards to support secure org-level data.

Deep Learning/AI Research Intern Jun 2024 – Aug 2024

UIC Department of Electrical & Computer Engineering *Chicago, IL*

- Led a team of five to design and prototype *Vision Mamba*, a custom neural network achieving +30% accuracy over baselines while optimizing for low-latency, high-throughput inference on constrained edge devices, and completed specialized advanced HW/SW co-design training using hls4ml to synthesize PyTorch models into FPGA bitstreams.

Projects & Research

SAGE – Secure Agentic Governed Execution | *Python, FastAPI, PostgreSQL, Harness Engineering* [GitHub]

- Independently authoring a research-driven architecture to identify & enforce common governance invariants across AI-executed work & reusable AI artifacts, including persistent context, authorization, and human-in-the-loop controls.

Machine Learning, Chess & Mathematics Research | *UIC MSCS MURL (MATH 496), Python, Machine Learning*

- Conducting faculty-supervised research applying machine learning and mathematical analysis to chess, with a focus on engine-guided evaluation, algorithmic decision-making, and iterative training of a custom model using Stockfish.

PocketFilter – FPGA Accelerator for Real-Time DNA Target Detection | *FPGA, Vitis HLS, PYNQ, DTW*

- Developing a battery-friendly, GPU-free FPGA accelerator on the AMD/Xilinx AUP-ZU3 for real-time nanopore DNA target detection through a faculty-sponsored Senior Design project, using Dynamic Time Warping, Python-backed golden-model validation, Vitis HLS, and throughput/power benchmarking for field-deployable, portable sequencing.

Technical Skills

Credentials: Anthropic: Claude Certified Architect – Foundations (In Progress, Oct 2026); AWS: Agentic AI Badge

Languages: Python, C++, C, JavaScript, TypeScript, Java, SQL, PHP, Bash, ARM/MIPS Assembly, SystemVerilog

Technologies: AWS, GCP, Docker, Linux, CI/CD, REST APIs, React, Node.js, PyTorch, Terraform, Vivado, FPGA